

Authentication gateway

The authentication gateways in many VPNs can also be effectively used as a firewall. If it's capable, it may be better to use simply the VPN's firewall

Setup help

Setting up a VPN is often very vendor-specific. However, many have web-based user interfaces that allow simple installation guides and wizards

VPN 101

Basics of setting up for Telecommuting

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Imagine you've finally managed to take a day off from work, and are at a long-awaited lunch with your friends. All of a sudden your boss calls you, and says that you have look at some data stored on the office network and mail a short summary to him. Right. Now. How do you view data stored on your office computer, or on a fileserver in office? One option is by keeping such data shared on a cloud-based service like Dropbox, but they can be quite insecure and vulnerable. In such a scenario, what you really need is a Virtual Private Network (VPN).

A VPN is a technology for using the internet or another intermediate network to connect computers to isolated remote computer networks that would otherwise be inaccessible. A VPN can provide varying levels of security so that data sent through the VPN connection can remain isolated from other access points on the intermediate network, either through the use of a dedicated connection from "one end" of the VPN to the other, or through encryption. VPNs are also capable of connecting multiple networks together, apart from connecting individual users to a remote network.

If you're part of a small corporate setup that has employees who need to frequently work while travelling, or work-from-home, you need to invest in a solution that lets such employees connect to the office network and access data and resources on it via a secure connection. In case you have a really small setup where only one or two people need to connect



VPNs can greatly improve your employees' accessibility and portability

to a single company computer, you can use remote desktop access software like GoToMyPC or LogMeIn. However, if you need multiple remote connections, a full VPN may be a better idea.

Protocols to be followed

Any VPN can follow one of several protocols available, or can even opt for a "hybrid protocol", combining two or more elements from two different protocols. We'll look at three of the biggest, and most commonly used protocols for VPNs today:

1. **PPTP:** Point-to-Point Tunnelling Protocol (PPTP) was first implemented by Microsoft in Windows 95, and VPNs for it have been around ever since. All Windows versions since Windows 95 have default support for PPTP, as do Mac OS X and some smartphones and VPN devices. However, this protocol

has severe security issues, and hence its use is gradually being phased out everywhere. So, while PPTP is still available, and will mostly be available for your device, choosing other more secure protocols may be a better choice.

2. **IPSec:** Internet Protocol Security (IPSec) was created by the Internet Engineering Task Force (IETF) and is a standard that encrypts all network traffic at a low level. IPSec is one of the most commonly used protocols by many vendors, like Cisco, Juniper, and Microsoft, and even by open-source projects, like KAME and Openswan, as the foundation for their VPN software. The greatest advantage of IPSec is that it requires all remote users to go through two levels of verification before they are granted access – a basic authentication to connect to the